ALI ERTAZA

Aspiring Data Engineer

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PROFESSIONAL SUMMARY

Final-year Software Engineering graduate (CGPA: 3.79) with hands-on experience building end-to-end data pipelines, Lakehouse architectures, and big data processing systems. Proficient in Apache Spark, Hadoop, Kafka, Delta Lake, Databricks, and AWS data services (Glue, Athena, Redshift, EMR, S3) through multiple self-driven and academy projects. Strong SQL, Python, and data modeling foundation, complemented by BI/visualization experience using Power BI and Tableau. Seeking a Data Engineer role to contribute to scalable data infrastructure and analytics solutions.

TECHNICAL SKILLS

Languages: Python, SQL, Java (basic), Kotlin

Big Data & Streaming: Apache Spark (PySpark), Apache Hadoop, Apache Kafka, Apache Airflow, Hive, Delta Lake

Cloud (AWS): S3, Glue, Athena, Redshift, EMR, IAM, EC2, VPC, Cloud9, CloudFormation

Cloud (Databricks): Databricks Workflows, Unity Catalog, Lakehouse (Bronze/Silver/Gold) Architecture

Databases: PostgreSQL, MySQL, MongoDB

BI & Visualization: Power BI, Tableau, DAX, KPI Dashboards, Data Modeling

Other Tools: Docker, Git/GitHub, Django, REST APIs, Linux/Bash

PROJECTS

CityFlow — Production-Grade Data Lakehouse | *Databricks, PySpark, Delta Lake, AWS S3, Unity Catalog* GitHub Repo

- Designed and built a medallion architecture (Bronze → Silver → Gold) data lakehouse processing 10.9M+ NYC Taxi records using Apache Spark 4.1.0 and Delta Lake on Databricks
- Ingested raw data from AWS S3 into the Bronze layer, performed cleaning/validation in the Silver layer (10.8M clean records), and built business-level aggregations in the Gold layer (tracking $168M+ revenue)
- Integrated real-time air quality streaming from the Open-Meteo API, implementing PM2.5/PM10/Ozone trend analysis
- Automated the end-to-end pipeline using Databricks Workflows with scheduled daily runs, achieving a 2 min 12 sec runtime, and implemented Unity Catalog for governance with credentials secured via Databricks Secrets

Real-Time Credit Card Fraud Detection Pipeline | *PySpark, Hadoop, Kafka, Scikit-learn*

- Built an end-to-end big data pipeline on Apache Spark with Hadoop/JVM configuration, processing 284,807+ transaction records (Kaggle Credit Card Fraud dataset)
- Performed distributed EDA and preprocessing using Spark DataFrames, applying SMOTE to address severe class imbalance (0.17% fraud)
- Implemented a real-time streaming pipeline using Apache Kafka for transaction ingestion and batch scoring
- Trained and evaluated Logistic Regression, Random Forest, and Gradient Boosted Trees models with ROC/PR curve analysis and feature importance ranking

AWS Cloud Data Lake & Analytics (AWS Academy Labs) | *S3, Glue, Athena, Redshift, EMR/Hive, IAM, VPC, EC2*

- Built a data lake using Amazon S3, transforming CSV datasets to Apache Parquet for optimized storage and querying, and configured AWS Glue crawlers to infer schema and build a Data Catalog
- Performed ETL using AWS Glue and queried datasets via Amazon Athena, including creating views for simplified analysis
- Deployed and queried an Amazon Redshift cluster for large-scale columnar data storage and analysis of a music ticket sales dataset
- Processed large weblog datasets (impression/click logs) using Amazon EMR and Apache Hive to identify ad click-through trends
- Configured IAM users, groups, and least-privilege policies, and set up a custom VPC with subnets, security groups, and an EC2 web server

IMDb Movies & TV Shows ETL Pipeline | *Python (Pandas), PostgreSQL, OOP* GitHub Repo

- Built a fully automated, modular ETL pipeline for the IMDb dataset using Python (Pandas) and PostgreSQL, following OOP principles for reusability across datasets
- Designed a star schema data warehouse model (movies, directors, actors, genres) with many-to-many relationships via junction tables
- Wrote 15+ analytical SQL queries (ratings, votes, runtime, genre trends, top actors/directors) with logging, progress tracking, and CSV export, plus full CRUD support

BI Dashboards & KPI Analytics | *Power BI, Tableau, DAX*

- Built interactive dashboards and KPI reports across multiple domains (sales analytics, fraud detection, Netflix/IMDb content analytics) using Power BI and Tableau
- Designed DAX measures and calculated columns for dynamic KPI tracking (revenue trends, churn, conversion rates, fraud patterns), with star-schema data modeling for optimized performance
- Delivered multiple paid client projects covering end-to-end data visualization workflows, from raw data to actionable business dashboards

EDUCATION

BS Software Engineering COMSATS University Islamabad, Sahiwal Campus, Pakistan
Sep 2022 – Sep 2026 | CGPA: 3.79 / 4.00